

perturbation method

$$x^2 + \epsilon x - 1 = 0 \quad \text{where } \epsilon \ll 1$$

$$\text{unperturbed } (\epsilon = 0): x^2 - 1 = 0 \Rightarrow x_0 = \pm 1$$

perturbation Assume $x = x_0 + \epsilon x_1$

Substituting into $x^2 + \epsilon x - 1 = 0$ yields

$$(\pm 1 + \epsilon x_1)^2 + \epsilon(\pm 1 + \epsilon x_1) - 1 = 0$$

Neglecting higher order terms

Root's are

$$x = 1 - \frac{\epsilon}{2} + \frac{\epsilon^2}{8} \quad \& \quad x = -1 - \frac{\epsilon}{2} - \frac{\epsilon^2}{8}$$

Take $\epsilon = 0.1$ & find the root-

_____ = 0.05 to 0.1 0.05, 0.06, 0.07, 0.08, 0.09, 0.1

$$x = 1 - \frac{0.1}{2} + \frac{0.1^2}{8}$$

$$= \boxed{}$$

$$\& x = -1 - \frac{0.1}{2} - \frac{0.1^2}{8}$$

$$= \boxed{}$$

Common Examples of Perturbation for Roots

- **Quadratic Equation ($x^2 + \epsilon x - 1 = 0$):**
 - **Unperturbed ($\epsilon = 0$):** $x^2 - 1 = 0 \Rightarrow x_0 = \pm 1$.
 - **Perturbation:** Assume $x = x_0 + \epsilon x_1$. Substituting into $x^2 + \epsilon x - 1 = 0$ yields $(\pm 1 + \epsilon x_1)^2 + \epsilon(\pm 1 + \epsilon x_1) - 1 = 0$.
 - **Result:** The roots are $x \approx 1 - \frac{\epsilon}{2} + \frac{\epsilon^2}{8}$ and $x \approx -1 - \frac{\epsilon}{2} - \frac{\epsilon^2}{8}$.
- **Cubic Polynomial ($x^3 - 3x^2 + 2x + \epsilon = 0$):**
 - **Unperturbed ($\epsilon = 0$):** $x(x-1)(x-2) = 0 \Rightarrow x_0 = 0, 1, 2$.
 - **Perturbation:** For the root near $x = 0$, assume $x \approx a_1\epsilon + a_2\epsilon^2$. Substituting and collecting powers of ϵ solves for a_1, a_2 , etc., approximating the shifted roots.
- **Algebraic Equation with Singular Perturbation ($x^3 - x - \epsilon = 0$):**
 - If a perturbation changes the degree of the equation or makes the leading order term vanish, it is a singular perturbation.
 - **Example:** For $\epsilon x^3 - x + 1 = 0$, the standard expansion fails. Instead, rescaling x (e.g., $x = \epsilon^{-1/2}y$) is necessary to find the behavior of roots.
- **Root Splitting (Multiple Roots):**
 - If the unperturbed equation has a double root, e.g., $(x-1)^2 - \epsilon = 0$, the roots are $x = 1 \pm \sqrt{\epsilon}$. This represents a case where the perturbation expansion requires fractional powers of ϵ .

